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Dataset: Loan Prediction Problem
Github Repo: https://github.com/dhruvsharaf18/loan-prediction

## Project Definition

Bank loan decisions depend on applicant information, but it is rarely clear which information matters most or how much. This project examines a dataset of 614 loan applications and performs data cleaning, exploratory analysis, and visualization on the dataset to find which applicant characteristics are associated with loan approval, and to assess how reliable those associations are given the data's quality problems. The goal is descriptive analysis. It does not build a production lending system, and its findings should not be used to make real credit decisions.

The project succeeds if a reader can (a) reproduce every cleaning step from the notebook, (b) see the justification for each decision, and (c) understand which findings are robust and which are fragile. Plot count is not a success criterion.

## Dataset Use Case

What the dataset represents: Each row is one loan application. It records the applicant's gender, marital status, dependents, education, employment type, income, co-applicant income, requested loan amount and term, credit history, property location, and whether the loan was approved. It is a classification-style dataset: the target is binary (Y or N).

It is used to study the relationship between applicant features and approval, and to demonstrate data preprocessing: handling missing values, duplicates, data types, and outliers. The dataset is small and its origin is undocumented. Results should be read as illustrative, not representative of real lending.

## Planned Procedure

Data loading and inspection: Load train.csv with pandas. Before changing anything, record the shape, data types, missing-value counts per column, duplicate rows, and duplicate Loan_IDs. Keeping these raw counts lets every later step be traced.

Duplicates: Check for duplicate rows and duplicate Loan_IDs. If none are found, report that. A null result is still a result.

Data types: Convert Dependents to an ordered category, with "3+" handled consistently. Convert Credit_History to a category so it isn't treated as a continuous number. Encode Loan_Status as 1 (Y) and 0 (N) for rate calculations.

Outliers: Flag values outside the 1.5 × IQR range for ApplicantIncome, CoapplicantIncome, and LoanAmount. Keep them by default, since a high income is a real applicant, not an error. Use log transforms for plotting and modeling-style comparisons. Remove a value only if it is impossible, such as a zero LoanAmount, and document every removal.

Missing values:

| Columns | Planned treatment | Justification |
|---|---|---|
| Gender, Married, Dependents, Self_Employed | Mode | Low missingness. Mode keeps the category set intact. It pushes records toward the majority category, so note that. |
| LoanAmount, Loan_Amount_Term | Median | These are skewed, and the median is robust to outliers. |
| Credit_History | Keep missing as a separate "Unknown" category | Filling with 1 would assume the strongest signal for applicants with no recorded history. A missing value may itself be informative. |

## Visualization

| Fig. | Plot | Type | Question it answers |
|---|---|---|---|
| 1 | Loan_Status distribution | Countplot | How imbalanced is the target? |
| 2 | ApplicantIncome, raw vs log | Two histograms | Is income skewed, and does log reduce the skew? |
| 3 | LoanAmount by Property_Area | Boxplot | Do loan sizes differ by location? |
| 4 | Approval rate by Credit_History | Bar chart | How strong is the credit-history association? |

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|---|---|---|---|
| 5 | Approval rate by Property_Area | Bar chart | Does location relate to approval? |
| 6 | Correlation among numeric features | Heatmap | Are any numeric features redundant? |
| 7 | ApplicantIncome vs LoanAmount, colored by status | Scatter plot | Does income alone separate approvals from rejections? |

## Expected Output

- Figure 1: Y is expected to be the majority class. The exact proportion will be reported from the data.
- Figure 2: Income is expected to be right-skewed, with log transformation producing a more symmetric shape. This is typical of income data in general.
- Figure 3: No strong prior. The plot tests whether location is associated with loan size.
- Figure 4: Credit history is expected to be the strongest of the categorical signals, because it is the standard lending criterion.
- Figure 5: Expected to be weaker than Figure 4. Any location effect is likely to be modest.
- Figure 6: Expected moderate correlation between LoanAmount and income, and weak correlation between the two income columns.
- Figure 7: Expected substantial overlap between approved and rejected applications, meaning income alone does not separate them cleanly.

## Python Libraries and Tools Used

- pandas: loading, cleaning, and aggregation
- NumPy: numerical operations and log transforms
- Matplotlib: base plotting
- Seaborn: statistical plots (countplots, boxplots, heatmaps)
- Jupyter Notebook: development and presentation of analysis
- Git and GitHub: version control and submission